Scientific Research Methodology
3. Indicators of publication quality

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Case studies

Consider the following papers:

- https://doi.org/10.14445/22315373/IJMTT-V69I5P502
- https://doi.org/10.4007/annals.2024.200.1.1
- https://doi.org/10.1016/j.jcp.2023.112083
- https://arxiv.org/html/2206.01179v12

How do we assess the credibility of a scientific publication?

What to check

- Who is the author? What is their track record? What is their academic affiliation?
- In which journal or book are the results published?
  - is the journal established and reputable?
  - who are the members of its editorial board?
  - is the book self-published or issued by an academic publisher?
  - has it undergone peer review, or is it simply a document posted online?
- Does the paper have a substantial reference list?
- Is the paper cited by other authors in the field?

Criteria for classifying journals

- Inclusion in established databases: Web of Science, ISI, MathSciNet
- Inclusion in free databases does not establish a high level of reliability
  - Example: any publication (even a non-academic one) can be indexed in Google Scholar!
- Journal ranking: comparison with other journals, citations over the last x years, self-citations, etc.
  - Example: `https://www.scimagojr.com/`
  - See the Wikipedia discussion: link
  - CiteScore: link

Requirements for a quality paper

- a paper assessed anonymously by other researchers: peer review
- Open Access vs. subscription
- Consider financial incentives: some journals use a fully Open Access model, with a fee for every publication! How can we determine whether the prospect of receiving a publication fee influences acceptance?

Assessing the quality of a paper

- in which journal is it published?
- has it passed peer review? Is it formally published?
- how many citations does it have? Who cites it?

Similar criteria apply when assessing the quality of a researcher's results

Why are such assessments carried out?

- obtaining an academic position or promotion
- obtaining research grants